

# Alexandre Bismuth

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## Education

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**Harvard University**, Cambridge MA, USA

2026 – 2027

MS Data Science

- **Coursework:** *Data Science, Machine Learning Topics, Sparse Inference, Efficient Deep Learning, Sensorimotor Learning.*
- **Anne Dias Grffin and Kenneth C. Griffin Fellow**

**University of Oxford**, Oxford, UK

2025 – 2026

MSc in Advanced Computer Science (AI Specialisation)

- **Coursework:** *Geometric Deep Learning, Uncertainty in Deep Learning, Computer Vision, Computational Medicine, Computational Game Theory, Algorithmic Foundations of Collective Decision-Making, Database Systems Implementation.*

**École Polytechnique**, Paris, France

2022 – 2025

BSc in Mathematics & Computer Science • GPA 3.8/4.0 (*Cum Laude*)

- **Selected Coursework:** *Linear Algebra, Analysis, Probability, Statistics, Stochastic Processes, Machine Learning, Numerical Analysis, Web Dev, Algorithms, Complexity, Computer Proofs, Formal Languages, Object-oriented Prog, Computer Architecture, Concurrency, Graphics.*
- **International Honors Program, Stanford University:** Coursework in *Tech Entrepreneurship* and *Artificial Intelligence* • GPA 3.7
- **Visiting student, University of Oxford:** Coursework in *Theories of Deep Learning, Graph Representation Learning, Graph Theory, Information Theory, Ethics of AI* • GPA 4.0

**École Jeannine Manuel**, Paris, France

2015 - 2022

French International Baccalaureate, American Section • awarded with highest honours (17.5/20)

- **Majors:** *Mathematics, Physics & Chemistry, Computer Science* (20/20 for each) — *Further Mathematics* elective

## Experience

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**Master's Thesis**, British Open-Ended Learning & Discovery Lab, UK

Apr 26 – Aug 26

- Building large market foundation models for high-frequency trading co-supervised by Prof Ani Calinescu and Prof Jakob Foerster
- Post-trained a Mamba-3 SSM with GANs and reinforcement learning (GRPO, EGGROLL) to predict Limit Order Book messages

**Bachelor Thesis**, University of Oxford, UK

Jan 25 – Mar 25

- Wrote "Optimising Gunshot Detection in Tropical Forests for Wildlife Protection with TinyML" supervised by Prof Alex Rogers
- Developed a SoTA ML Arduino pipeline with 256kB RAM and 1MB memory via model compression and knowledge distillation

**Software Development Engineer Intern**, Amazon, Spain

Jul 24 – Sep 24

- Designed and implemented the migration of Amazon Charts' backend architecture over three months, reducing costs by \$10k/year
- Shipped features for Amazon Books, deploying 'Best Sellers' reports to South Africa/Poland and creating new targeting filters

**Research Assistant**, Stanford University, USA

Jul 24 – Sep 24

- Conducted part-time research in reinforcement learning and state extrapolation with Prof Itai Ashlagi on kidney exchange policies
- Applied Q-learning to train a customisable agent queuing system with a large state space and infer optimal policies for new states

## Leadership

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**Duo Conseil X-HEC – Treasurer**

2023 – 2025

- Oversaw and participated in operations of a 50-member consulting society as board member – generated \$60k of ARR

**X'Srael, The Polytechnique Entrepreneurship Association - President**

2022 – 2024

- Planned a fully-funded TechTrip in February 2023, allowing 42 students to discover the start-up nation – raised \$50k

## Skills & Interests

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**Programming** Python, C++, C, Java, Kotlin, JavaScript, Scala, Haskell, SQL, Rocq || PyTorch, TensorFlow, AWS, HPCs

**Languages** French (Native), English (Fluent - 116/120 TOEFL IBT), Spanish (Fluent)

**Interests** Music: Singing, Piano, Guitar || Sports: Endurance Running, Volleyball, Rowing, Padel, Rugby